

OPERA NUMERORUM

NS Tower -- Path A: Phase 102
NS_Zerolnit_Pointwise_PROVED via Open-Set Measure Argument

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SECTION 1: THEOREM PROVED

NS_Zerolnit_Pointwise_PROVED (0 sorry, classical trio):

HYPOTHESIS: $\forall t : \text{EuclideanSpace } \mathbb{R}^3 \rightarrow \text{EuclideanSpace } \mathbb{R}^3$ continuous integral x , $\|\forall t \ x\|^2 d_{\text{haar}} = 0$ CONCLUSION: $\text{forall } x, \forall t \ x = 0$

SECTION 2: PROOF (3 STEPS, 0 SORRY)

Step	Content	Lean API
1	integral nonneg = 0 $\rightarrow$ ae zero $\ \forall t \ x\ ^2 \geq 0$ and integral = 0 $\rightarrow \ \forall t \ x\ ^2 = 0$ a.e. [haar]	integral_eq_zero_iff_of_nonneg_ae sq_nonneg, AEMeasurable
2	Open-set measure argument: Assume $\ \forall t \ x_0\ ^2 > 0$ for some x_0 . $U = \{y \mid \ \forall t \ y\ ^2 > 0\}$ is open (preimage of $(0, \infty)$ under continuous function). $x_0 \in U \rightarrow U \text{ nonempty} \rightarrow \text{haar}(U) > 0$ [IsOpenPosMeasure: Haar is positive on open sets]. $U \subset \{y \mid \ \forall t \ y\ ^2 \neq 0\} \rightarrow \text{haar}(U) = 0$ [from a.e. = 0 via ae_iff + measure_mono_null]. $0 < \text{haar}(U) = 0$: CONTRADICTION. Hence forall x , $\ \forall t \ x\ ^2 = 0$.	IsOpenPosMeasure.measure_pos_of_nonempty_open measure_mono_null, ae_iff (Haar on $\mathbb{R}^3$: positive on open sets)
3	$\ \forall t \ x\ ^2 = 0 \rightarrow \forall t \ x = 0$ $\ \forall t \ x\ \geq 0$ and $\ \forall t \ x\ ^2 = 0$ $\rightarrow \ \forall t \ x\ = 0 \rightarrow \forall t \ x = 0$	nlarith [norm_nonneg, sq_nonneg] norm_eq_zero

SECTION 3: KEY MATHEMATICAL INSIGHT — IsOpenPosMeasure

The Haar measure on EuclideanSpace $\mathbb{R}$ (Fin 3) ($= \mathbb{R}^3$) is IsOpenPosMeasure: every nonempty open set has strictly positive measure. This is because Haar measure is locally finite (finite on compact sets) and translation-invariant on $\mathbb{R}^3$. Any open ball $B(x, \epsilon)$ has positive measure ($= c * \epsilon^3$ for the standard Lebesgue normalization). This IsOpenPosMeasure property, combined with continuity, is the key tool that lifts ' $f = 0$ a.e.' to ' $f = 0$ everywhere' for continuous functions.

Lean 4 type class chain: EuclideanSpace $\mathbb{R}$ (Fin 3) = PiLp 2 (fun _ : Fin 3 => $\mathbb{R}$)
LocallyCompactSpace (EuclideanSpace $\mathbb{R}$ (Fin 3)) [standard Mathlib instance]
IsTopologicalGroup (EuclideanSpace $\mathbb{R}$ (Fin 3)) [additive group structure] Measure.haar :
Measure (EuclideanSpace $\mathbb{R}$ (Fin 3)) [Haar measure] IsOpenPosMeasure Measure.haar [locally compact groups]

SECTION 4: GAP TABLE -- Phase 102, 6 deps (7 $\rightarrow$ 6)

#	Named Open Def	ETA	Status
1	NS_ESSRescaleNS_OPEN	2-4 weeks	OPEN (NEXT)

2	NS_Carleman_SmoothApprox_OPEN	3-6 weeks	OPEN
3	NS_BlowupConcentration_OPEN	2-3 months	OPEN
4	NS_Carleman_LimitPass_OPEN	2-4 months	OPEN
5	NS_CarlemanHeat_OPEN	3-6 months	CRITICAL
6	NS_CarlemanDriftAbsorption_OPEN	after heat	BLOCKED

SECTION 5: INTEGRITY

Repository: DavidFox998/navier-stokes New file:

Towers/NS/NSPhase102ZeroInitPointwise.lean Lean toolchain: v4.12.0 | Mathlib: v4.12.0 CMI
rules: 0 sorry, 0 axiom keyword, named open defs only Cumulative:

95(7)->98(10)->99(8)->100(8)->101(7)->102(6) M7 manifest: FROZEN (5b80b84d) -- m1..m6.out
unchanged #print axioms NS_M6_CLOSED_v102 -> {propext, Classical.choice, Quot.sound}